Inverse Trigonometric Functions

Only One Option Correct

1. Considering the principal values of the inverse trigonometric functions, $\sin^{-1}\left(\frac{\sqrt{3}}{2}x+\frac{1}{2}\sqrt{1-x^2}\right)$, $-\frac{1}{2}<x<\frac{1}{\sqrt{2}}$, is equal to

(a) $\frac{-5\pi}{6}-\sin^{-1}x$ (b) $\frac{\pi}{4}+\sin^{-1}x$

(c) $\frac{5\pi}{6}-\sin^{-1}x$ (d) $\frac{\pi}{6}+\sin^{-1}x$

(Main 4th April 1st Shift 2025)

2. The sum of the infinite series

$\cot^{-1}\left(\frac{7}{4}\right)+\cot^{-1}\left(\frac{19}{4}\right)+\cot^{-1}\left(\frac{39}{4}\right)+\cot^{-1}\left(\frac{67}{4}\right)+\ldots$ is :

(a) $\frac{\pi}{2}+\cot^{-1}\left(\frac{1}{2}\right)$ (b) $\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{2}\right)$

(c) $\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{2}\right)$ (d) $\frac{\pi}{2}-\cot^{-1}\left(\frac{1}{2}\right)$

(Main 4th April 2nd Shift 2025)

3. Let the domains of the functions $f(x) = \log_4 \log_3 \log_7(8-\log_2(x^2+4x+5))$ and $g(x)=\sin^{-1}\left(\frac{7x+10}{x-2}\right)$ be (α, β) and $[\gamma, \delta]$, respectively. Then $\alpha^2+\beta^2+\gamma^2+\delta^2$ is equal to :

(a) 13 (b) 16 (c) 15 (d) 14

(Main 4th April 2nd Shift 2025)

4. The value of

$$\cot^{-1}\left(\frac{\sqrt{1+\tan^2(2)}-1}{\tan(2)}\right)-\cot^{-1}\left(\frac{\sqrt{1+\tan^2\left(\frac{1}{2}\right)}+1}{\tan\left(\frac{1}{2}\right)}\right)$$

is equal to

(a) $\pi+\frac{3}{2}$ (b) $\pi-\frac{5}{4}$ (c) $\pi-\frac{3}{2}$ (d) $\pi+\frac{5}{2}$

(Main 8th April 2nd Shift 2025)

5. Using the principal values of the inverse trigonometric functions, the sum of the maximum and the minimum values of $16((\sec^{-1}x)^2+(\operatorname{cosec}^{-1}x)^2)$ is :

(a) $22\pi^2$ (b) $24\pi^2$ (c) $18\pi^2$ (d) $31\pi^2$

(Main 22nd Jan 1st Shift 2025)

6. If $\frac{\pi}{2}\le x\le\frac{3\pi}{4}$, then $\cos^{-1}\left(\frac{12}{13}\cos x+\frac{5}{13}\sin x\right)$ is equal to

(a) $x-\tan^{-1}\frac{5}{12}$ (b) $x+\tan^{-1}\frac{5}{12}$

(c) $x+\tan^{-1}\frac{4}{5}$ (d) $x-\tan^{-1}\frac{4}{3}$

(Main 23rd Jan 1st Shift 2025)

7. If $\alpha>\beta>\gamma>0$, then the expression $\cot^{-1}\left\{\beta+\frac{(1+\beta^2)}{(\alpha-\beta)}\right\}+\cot^{-1}\left\{\gamma+\frac{(1+\gamma^2)}{(\beta-\gamma)}\right\}+\cot^{-1}\left\{\alpha+\frac{(1+\alpha^2)}{(\gamma-\alpha)}\right\}$ is equal to:

(a) 3π (b) 0

(c) $\frac{\pi}{2}-(\alpha+\beta+\gamma)$ (d) π

(Main 24th Jan 2nd Shift 2025)

8. $\cos\left(\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{33}{65}\right)$ is equal to :

(a) 1 (b) 0 (c) 32/65 (d) 33/65

(Main 28th Jan 1st Shift 2025)

9. Let $[x]$ denote the greatest integer less than or equal to x. Then the domain of $f(x) = \sec^{-1}(2[x]+1)$ is :

(a) $(-\infty, \infty)$ (b) $(-\infty, \infty)-\{0\}$

(c) $(-\infty, -1]\cup[0, \infty)$ (d) $(-\infty, -1]\cup[1, \infty)$

(Main 28th Jan 2nd Shift 2025)

10. The total number of real solutions of the equation

$\theta=\tan^{-1}(2\tan\theta)-\frac{1}{2}\sin^{-1}\left(\frac{6\tan\theta}{9+\tan^2\theta}\right)$ is

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\tan^{-1}x$ assume values in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ and $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, respectively.)

(a) 1 (b) 2 (c) 3 (d) 5

(Advanced 2025)

11. If the domain of the function

$\sin^{-1}\left(\frac{3x-22}{2x-19}\right)+\log_e\left(\frac{3x^2-8x+5}{x^2-3x-10}\right)$ is $(\alpha, \beta]$, then $3\alpha+10\beta$ is equal to

(a) 95 (b) 97 (c) 98 (d) 100

(Main 4th April 1st Shift 2024)

12. Given that the inverse trigonometric function assumes principal values only. Let x, y be any two real numbers in $[-1, 1]$ such that $\cos^{-1}x-\sin^{-1}y=\alpha$, $\frac{-\pi}{2}\le\alpha\le\pi$. Then, the minimum value of $x^2+y^2+2xy\sin\alpha$ is

(a) -1 (b) 0 (c) $\frac{1}{2}$ (d) $\frac{-1}{2}$

(Main 4th April 2nd Shift 2024)

13. If the domain of the function $f(x)=\sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is $\mathbf{R}-(\alpha, \beta)$, then $12\alpha\beta$ is equal to :

(a) 24 (b) 32 (c) 40 (d) 36

(Main 9th April 1st Shift 2024)

14. Considering only the principal values of inverse trigonometric functions, the number of positive real values of x satisfying $\tan^{-1}(x)+\tan^{-1}(2x)=\frac{\pi}{4}$ is :
(a) 1 (b) more than 2
(c) 2 (d) 0
(Main 27th Jan 2nd Shift 2024)

15. Let $x=\frac{m}{n}$ (m, n are co-prime natural numbers) be a solution of the equation $\cos(2\sin^{-1}x)=\frac{1}{9}$ and let α, β ($\alpha>\beta$) be the roots of the equation $mx^2-nx-m+n=0$. Then the point (α, β) lies on the line
(a) $3x-2y=-2$ (b) $5x-8y=-9$
(c) $3x+2y=2$ (d) $5x+8y=9$
(Main 29th Jan 2nd Shift 2024)

16. If the domain of the function
$f(x)=\cos^{-1}\left(\frac{2-|x|}{4}\right)+\{\log_e(3-x)\}^{-1}$ is $[-\alpha, \beta)-\{\gamma\}$, then $\alpha+\beta+\gamma$ is equal to
(a) 9 (b) 8 (c) 11 (d) 12
(Main 30th Jan 1st Shift 2024)

17. If the domain of the function
$f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ is $(\alpha, \beta]$, then the value of $5\beta-4\alpha$ is equal to
(a) 12 (b) 10 (c) 11 (d) 9
(Main 30th Jan 2nd Shift 2024)

18. For $\alpha, \beta, \gamma \neq 0$, if $\sin^{-1}\alpha+\sin^{-1}\beta+\sin^{-1}\gamma=\pi$ and $(\alpha+\beta+\gamma)(\alpha-\gamma+\beta)=3\alpha\beta$, then γ equals
(a) $\frac{\sqrt{3}}{2}$ (b) $\sqrt{3}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{\sqrt{3}-1}{2\sqrt{2}}$
(Main 31st Jan 1st Shift 2024)

19. If $a=\sin^{-1}(\sin(5))$ and $b=\cos^{-1}(\cos(5))$, then a^2+b^2 is equal to
(a) 25 (b) $8\pi^2-40\pi+50$
(c) $4\pi^2-20\pi+50$ (d) $4\pi^2+25$
(Main 31st Jan 2nd Shift 2024)

20. Considering only the principal values of the inverse trigonometric functions, the value of
$\tan\left(\sin^{-1}\left(\frac{3}{5}\right)-2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$ is
(a) $\frac{7}{24}$ (b) $\frac{-7}{24}$ (c) $\frac{-5}{24}$ (d) $\frac{5}{24}$
(Advanced 2024)

21. Let D be the domain of the function $f(x)=\sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$. If the range of the function $g: D\to R$ defined by $g(x)=x-[x]$, ($[x]$ is the greatest integer function), is (α, β), then $\alpha^2+\frac{5}{\beta}$ is equal to
(a) 135 (b) 136 (c) 45 (d) 46
(Main 12th April 1st Shift 2023)

22. The range of $f(x)=4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is
(a) $[0, 2\pi)$ (b) $[0, \pi]$ (c) $[0, 2\pi]$ (d) $[0, \pi)$
(Main 13th April 2nd Shift 2023)

23. If the domain of the function $f(x)=\log_e(4x^2+11x+6)+\sin^{-1}(4x+3)+\cos^{-1}\left(\frac{10x+6}{3}\right)$ is $(\alpha, \beta]$, then $36|\alpha+\beta|$ is equal to
(a) 45 (b) 63 (c) 72 (d) 54
(Main 15th April 1st Shift 2023)

24. Let S be the set of all solutions of the equation
$\cos^{-1}(2x)-2\cos^{-1}(\sqrt{1-x^2})=\pi$, $x\in\left[-\frac{1}{2},\frac{1}{2}\right]$.
Then $\sum_{x\in S}2\sin^{-1}(x^2-1)$ is equal to
(a) 0 (b) $\frac{-2\pi}{3}$
(c) $\pi-2\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (d) $\pi-\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$
(Main 1st Feb 1st Shift 2023)

25. Let
$S=\left\{x\in\mathbf{R}: 0<x<1 \text{ and } 2\tan^{-1}\left(\frac{1-x}{1+x}\right)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right\}$.
If $n(S)$ denotes the number of elements in S then
(a) $n(S)=2$ and only one element in S is less than $\frac{1}{2}$.
(b) $n(S)=1$ and the element in S is less than $\frac{1}{2}$.
(c) $n(S)=1$ and the elements in S is more than $\frac{1}{2}$.
(d) $n(S)=0$
(Main 1st Feb 2nd Shift 2023)

26. $\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right)+\sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ is equal to
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{3}$
(Main 24th Jan 1st Shift 2023)

27. Let $a_1=1, a_2, a_3, a_4, ...$ be consecutive natural numbers.
Then $\tan^{-1}\left(\frac{1}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{1}{1+a_2a_3}\right)+....$
$+\tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$ is equal to
(a) $\cot^{-1}(2022)-\frac{\pi}{4}$ (b) $\frac{\pi}{4}-\cot^{-1}(2022)$
(c) $\tan^{-1}(2022)-\frac{\pi}{4}$ (d) $\frac{\pi}{4}-\tan^{-1}(2022)$
(Main 30th Jan 2nd Shift 2023)

28. If $\sin^{-1}\frac{\alpha}{17}+\cos^{-1}\frac{4}{5}-\tan^{-1}\frac{77}{36}=0$, $0<\alpha<13$, then $\sin^{-1}(\sin\alpha)+\cos^{-1}(\cos\alpha)$ is equal to
(a) $16-5\pi$ (b) 16
(c) 0 (d) π
(Main 31st Jan 1st Shift 2023)

29. Let $(a, b) \subset (0, 2\pi)$ be the largest interval for which $\sin^{-1}(\sin\theta) - \cos^{-1}(\sin\theta) > 0$, $\theta \in (0, 2\pi)$, holds. If $\alpha x^2 + \beta x + \sin^{-1}(x^2 - 6x + 10) + \cos^{-1}(x^2 - 6x + 10) = 0$ and $\alpha - \beta = b - a$, then α is equal to

(a) $\frac{\pi}{8}$ (b) $\frac{\pi}{48}$ (c) $\frac{\pi}{16}$ (d) $\frac{\pi}{12}$

(Main 31st Jan 2nd Shift 2023)

30. For any $y \in \mathbb{R}$, let $\cot^{-1}(y) \in (0, \pi)$ and $\tan^{-1}(y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Then the sum of all the solutions of the equation

$\tan^{-1}\left(\frac{6y}{9-y^2}\right) + \cot^{-1}\left(\frac{9-y^2}{6y}\right) = \frac{2\pi}{3}$ for $0 < |y| < 3$, is equal to

(a) $2\sqrt{3}-3$ (b) $3-2\sqrt{3}$

(c) $4\sqrt{3}-6$ (d) $6-4\sqrt{3}$ *(Advanced 2023)*

31. $\tan\left(2\tan^{-1}\frac{1}{5} + \sec^{-1}\frac{\sqrt{5}}{2} + 2\tan^{-1}\frac{1}{8}\right)$ is equal to

(a) 1 (b) 2 (c) $\frac{1}{4}$ (d) $\frac{5}{4}$

(Main 26th July 1st Shift 2022)

32. If $0 < x < \frac{1}{\sqrt{2}}$ and $\frac{\sin^{-1}x}{\alpha} = \frac{\cos^{-1}x}{\beta}$, then a value of $\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right)$ is

(a) $4\sqrt{(1-x^2)(1-2x^2)}$ (b) $4x\sqrt{(1-x^2)(1-2x^2)}$

(c) $2x\sqrt{(1-x^2)(1-4x^2)}$ (d) $4\sqrt{(1-x^2)(1-4x^2)}$

(Main 26th July 2nd Shift 2022)

33. Considering only the principal values of the inverse trigonometric functions, the domain of the function $f(x) = \cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ is

(a) $\left(-\infty, \frac{1}{4}\right]$ (b) $\left[-\frac{1}{4}, \infty\right)$

(c) $\left(-\frac{1}{3}, \infty\right)$ (d) $\left(-\infty, \frac{1}{3}\right]$

(Main 28th July 1st Shift 2022)

34. Considering the principal values of the inverse trigonometric functions, the sum of all the solutions of the equation $\cos^{-1}(x) - 2\sin^{-1}(x) = \cos^{-1}(2x)$ is equal to

(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

(Main 28th July 1st Shift 2022)

35. The domain of the function $f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ is:

(a) $[1, \infty)$ (b) $[-1, 2]$

(c) $[-1, \infty)$ (d) $(-\infty, 2]$

(Main 29th July 2nd Shift 2022)

36. The set of all values of k for which $(\tan^{-1}x)^3 + (\cot^{-1}x)^3 = k\pi^3$, $x \in R$, is the interval

(a) $\left[\frac{1}{32}, \frac{7}{8}\right)$ (b) $\left(\frac{1}{24}, \frac{13}{16}\right)$

(c) $\left[\frac{1}{48}, \frac{13}{16}\right]$ (d) $\left[\frac{1}{32}, \frac{9}{8}\right)$

(Main 24th June 1st Shift 2022)

37. Let $x*y = x^2 + y^3$ and $(x*1)*1 = x*(1*1)$

Then a value of $2\sin^{-1}\left(\frac{x^4+x^2-2}{x^4+x^2+2}\right)$ is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{6}$

(Main 24th June 2nd Shift 2022)

38. The value of $\tan^{-1}\left(\frac{\cos\left(\frac{15\pi}{4}\right)-1}{\sin\left(\frac{\pi}{4}\right)}\right)$ is equal to

(a) $-\frac{\pi}{4}$ (b) $-\frac{\pi}{8}$ (c) $-\frac{5\pi}{12}$ (d) $-\frac{4\pi}{9}$

(Main 25th June 2nd Shift 2022)

39. If the inverse trigonometric functions take principal values, then

$\cos^{-1}\left(\frac{3}{10}\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right) + \frac{2}{5}\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)\right)$ is equal to

(a) 0 (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

(Main 26th June 2nd Shift 2022)

40. $\sin^{-1}\left(\sin\frac{2\pi}{3}\right) + \cos^{-1}\left(\cos\frac{7\pi}{6}\right) + \tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ is equal to :

(a) $\frac{11\pi}{12}$ (b) $\frac{17\pi}{12}$ (c) $\frac{31\pi}{12}$ (d) $-\frac{3\pi}{4}$

(Main 27th June 1st Shift 2022)

41. The value of $\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ is

(a) $\frac{26}{25}$ (b) $\frac{25}{26}$ (c) $\frac{50}{51}$ (d) $\frac{52}{51}$

(Main 27th June 2nd Shift 2022)

42. The domain of the function $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$ is

(a) $R - \left\{-\frac{1}{2}, \frac{1}{2}\right\}$

(b) $(-\infty, -1] \cup [1, \infty) \cup \{0\}$

(c) $\left(-\infty, \frac{-1}{2}\right) \cup \left(\frac{1}{2}, \infty\right) \cup \{0\}$

(d) $\left(-\infty, \frac{-1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, \infty\right) \cup \{0\}$

(Main 29th June 1st Shift 2022)

43. $\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin(6)) - \tan^{-1}(\tan(12))$ is equal to (The inverse trigonometric functions takes the principal values)

(a) $3\pi + 1$ (b) $3\pi - 11$
(c) $4\pi - 11$ (d) $4\pi - 9$

(Main 1st Sept 2nd Shift 2021)

44. If $\sum_{r=1}^{50} \tan^{-1}\frac{1}{2r^2} = p$, then the value of $\tan p$ is

(a) 100 (b) $\frac{51}{50}$ (c) $\frac{50}{51}$ (d) $\frac{101}{102}$

(Main 26th Aug 2nd Shift 2021)

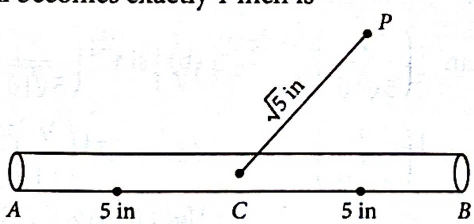
45. The domain of the function $\operatorname{cosec}^{-1}\left(\frac{1+x}{x}\right)$ is

(a) $\left[-\frac{1}{2}, \infty\right) - \{0\}$ (b) $\left(-1, -\frac{1}{2}\right] \cup (0, \infty)$

(c) $\left[-\frac{1}{2}, 0\right) \cup [1, \infty)$ (d) $\left(-\frac{1}{2}, \infty\right) - \{0\}$

(Main 26th Aug 2nd Shift 2021)

46. A 10 inches long pencil AB with mid point C and a small eraser P are placed on the horizontal top of a table such that $PC = \sqrt{5}$ inches and $\angle PCB = \tan^{-1}(2)$. The acute angle through which the pencil must be rotated about C so that the perpendicular distance between eraser and pencil becomes exactly 1 inch is



(a) $\tan^{-1}\left(\frac{3}{4}\right)$ (b) $\tan^{-1}\left(\frac{1}{2}\right)$

(c) $\tan^{-1}\left(\frac{4}{3}\right)$ (d) $\tan^{-1}(1)$

(Main 26th Aug 2nd Shift 2021)

47. If $(\sin^{-1}x)^2 - (\cos^{-1}x)^2 = a$; $0 < x < 1$, $a \neq 0$, then the value of $2x^2 - 1$ is

(a) $\cos\left(\frac{2a}{\pi}\right)$ (b) $\sin\left(\frac{2a}{\pi}\right)$

(c) $\cos\left(\frac{4a}{\pi}\right)$ (d) $\sin\left(\frac{4a}{\pi}\right)$

(Main 27th Aug 1st Shift 2021)

48. Let M and m respectively be the maximum and minimum values of the function $f(x) = \tan^{-1}(\sin x + \cos x)$ in $\left[0, \frac{\pi}{2}\right]$. Then the value of $\tan(M - m)$ is equal to

(a) $2-\sqrt{3}$ (b) $3-2\sqrt{2}$ (c) $3+2\sqrt{2}$ (d) $2+\sqrt{3}$

(Main 27th Aug 2nd Shift 2021)

49. The domain of the function

$$f(x) = \sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right) + \cos^{-1}\left(\frac{x-1}{x+1}\right) \text{ is}$$

(a) $\left[0, \frac{1}{2}\right]$ (b) $\left[0, \frac{1}{4}\right]$

(c) $\left[\frac{1}{4}, \frac{1}{2}\right] \cup \{0\}$ (d) $[-2, 0] \cup \left[\frac{1}{4}, \frac{1}{2}\right]$

(Main 31st Aug 2nd Shift 2021)

50. The number of real roots of the equation

$\tan^{-1}\sqrt{x(x+1)} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{4}$ is

(a) 2 (b) 0 (c) 4 (d) 1

(Main 20th July 1st Shift 2021)

51. The value of $\tan\left(2\tan^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right)\right)$ is equal to

(a) $\frac{151}{63}$ (b) $\frac{220}{21}$ (c) $\frac{-181}{69}$ (d) $\frac{-291}{76}$

(Main 20th July 2nd Shift 2021)

52. If the domain of the function $f(x) = \frac{\cos^{-1}\sqrt{x^2-x+1}}{\sqrt{\sin^{-1}\left(\frac{2x-1}{2}\right)}}$ is the interval $(\alpha, \beta]$, then $\alpha + \beta$ is equal to

(a) 1 (b) $\frac{1}{2}$ (c) 2 (d) $\frac{3}{2}$

(Main 22nd July 2nd Shift 2021)

53. Let $S_k = \sum_{r=1}^{k} \tan^{-1}\left(\frac{6^r}{2^{2r+1}+3^{2r+1}}\right)$.

Then $\lim_{k\to\infty} S_k$ is equal to

(a) $\cot^{-1}\left(\frac{3}{2}\right)$ (b) $\frac{\pi}{2}$

(c) $\tan^{-1}\left(\frac{3}{2}\right)$ (d) $\tan^{-1}(3)$

(Main 16th March 1st Shift 2021)

54. Given that the inverse trigonometric functions take principal values only. Then, the number of real values of x which satisfy $\sin^{-1}\left(\frac{3x}{5}\right) + \sin^{-1}\left(\frac{4x}{5}\right) = \sin^{-1}x$ is equal to

(a) 2 (b) 1 (c) 0 (d) 3

(Main 16th March 2nd Shift 2021)

55. If $\cot^{-1}(\alpha) = \cot^{-1}2 + \cot^{-1}8 + \cot^{-1}18 + \cot^{-1}32 + ...$ upto 100 terms, then α is
(a) 1.01 (b) 1.02 (c) 1.00 (d) 1.03
(Main 17th March 1st Shift 2021)

56. The sum of possible value of x for
$$\tan^{-1}(x+1)+\cot^{-1}\left(\frac{1}{x-1}\right)=\tan^{-1}\left(\frac{8}{31}\right) \text{ is}$$
(a) $-\frac{33}{4}$ (b) $-\frac{31}{4}$
(c) $-\frac{30}{4}$ (d) $-\frac{32}{4}$
(Main 17th March 1st Shift 2021)

57. The number of solutions of the equation
$\sin^{-1}\left[x^2+\frac{1}{3}\right]+\cos^{-1}\left[x^2-\frac{2}{3}\right]=x^2$, for $x \in [-1, 1]$ and $[x]$ denotes the greatest integer less than or equal to x, is
(a) 2 (b) 0 (c) 4 (d) infinite
(Main 17th March 2nd Shift 2021)

58. The real valued function $f(x)=\frac{\operatorname{cosec}^{-1}x}{\sqrt{x-[x]}}$, where $[x]$ denotes the greatest integer less than or equal to x, is defined for all x belonging to
(a) all integers except 0, –1, 1
(b) all reals except the interval [–1, 1]
(c) all reals except integers
(d) all non-integers except the interval [–1, 1]
(Main 18th March 1st Shift 2021)

59. A possible value of $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$ is
(a) $\frac{1}{2\sqrt{2}}$ (b) $2\sqrt{2}-1$ (c) $\sqrt{7}-1$ (d) $\frac{1}{\sqrt{7}}$
(Main 24th Feb 2nd Shift 2021)

60. $\operatorname{cosec}\left[2\cot^{-1}(5)+\cos^{-1}\left(\frac{4}{5}\right)\right]$ is equal to
(a) $\frac{56}{33}$ (b) $\frac{65}{56}$ (c) $\frac{65}{33}$ (d) $\frac{75}{56}$
(Main 25th Feb 2nd Shift 2021)

61. If $\frac{\sin^{-1}x}{a}=\frac{\cos^{-1}x}{b}=\frac{\tan^{-1}y}{c}$; $0 < x < 1$, then the value of $\cos\left(\frac{\pi c}{a+b}\right)$ is
(a) $\frac{1-y^2}{y\sqrt{y}}$ (b) $\frac{1-y^2}{1+y^2}$ (c) $1-y^2$ (d) $\frac{1-y^2}{2y}$
(Main 26th Feb 1st Shift 2021)

62. If $0 < a, b < 1$, and $\tan^{-1}a + \tan^{-1}b = \frac{\pi}{4}$, then the value of
$$(a+b)-\left(\frac{a^2+b^2}{2}\right)+\left(\frac{a^3+b^3}{3}\right)-\left(\frac{a^4+b^4}{4}\right)+.... \text{ is}$$
(a) e (b) e^2-1 (c) $\log_e 2$ (d) $\log_e\left(\frac{e}{2}\right)$
(Main 26th Feb 2nd Shift 2021)

63. The domain of the function $f(x)=\sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$ is $(-\infty, -a] \cup [a, \infty)$. Then a is equal to
(a) $\frac{\sqrt{17}}{2}$ (b) $\frac{\sqrt{17}-1}{2}$ (c) $\frac{1+\sqrt{17}}{2}$ (d) $\frac{\sqrt{17}}{2}+1$
(Main 2nd Sept 1st Shift 2020)

64. $2\pi-\left(\sin^{-1}\frac{4}{5}+\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{16}{65}\right)$ is equal to
(a) $\frac{\pi}{2}$ (b) $\frac{5\pi}{4}$ (c) $\frac{3\pi}{2}$ (d) $\frac{7\pi}{4}$
(Main 3rd Sept 1st Shift 2020)

65. If S is the sum of the first 10 terms of the series
$\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+\tan^{-1}\left(\frac{1}{21}\right)+...,$
then $\tan(S)$ is equal to
(a) $\frac{5}{6}$ (b) $\frac{5}{11}$ (c) $-\frac{6}{5}$ (d) $\frac{10}{11}$
(Main 5th Sept 1st Shift 2020)

66. If $\alpha=\cos^{-1}\left(\frac{3}{5}\right)$, $\beta=\tan^{-1}\left(\frac{1}{3}\right)$, where $0<\alpha,\beta<\frac{\pi}{2}$, then $\alpha-\beta$ is equal to
(a) $\tan^{-1}\left(\frac{9}{5\sqrt{10}}\right)$ (b) $\sin^{-1}\left(\frac{9}{5\sqrt{10}}\right)$
(c) $\cos^{-1}\left(\frac{9}{5\sqrt{10}}\right)$ (d) $\tan^{-1}\left(\frac{9}{14}\right)$
(Main 8th April 1st Shift 2019)

67. If $\cos^{-1}x-\cos^{-1}\frac{y}{2}=\alpha$, where $-1 \le x \le 1$, $-2 \le y \le 2$, $x \le \frac{y}{2}$, then for all x, y, $4x^2-4xy\cos\alpha+y^2$ is equal to
(a) $4\sin^2\alpha$ (b) $2\sin^2\alpha$
(c) $4\sin^2\alpha-2x^2y^2$ (d) $4\cos^2\alpha+2x^2y^2$
(Main 10th April 2nd Shift 2019)

68. The value of $\sin^{-1}\left(\frac{12}{13}\right)-\sin^{-1}\left(\frac{3}{5}\right)$ is equal to
(a) $\frac{\pi}{2}-\sin^{-1}\left(\frac{56}{65}\right)$ (b) $\pi-\sin^{-1}\left(\frac{63}{65}\right)$
(c) $\frac{\pi}{2}-\cos^{-1}\left(\frac{9}{65}\right)$ (d) $\pi-\cos^{-1}\left(\frac{33}{65}\right)$
(Main 12th April 1st Shift 2019)

69. If $\cos^{-1}\left(\frac{2}{3x}\right)+\cos^{-1}\left(\frac{3}{4x}\right)=\frac{\pi}{2}\left(x>\frac{3}{4}\right)$, then x is equal to

(a) $\frac{\sqrt{145}}{12}$ (b) $\frac{\sqrt{145}}{11}$ (c) $\frac{\sqrt{145}}{10}$ (d) $\frac{\sqrt{146}}{12}$

(Main 9th Jan 1st Shift 2019)

70. If $x = \sin^{-1}(\sin 10)$ and $y = \cos^{-1}(\cos 10)$, then $y - x$ is equal to

(a) 7π (b) 0 (c) π (d) 10

(Main 9th Jan 2nd Shift 2019)

71. The value of $\cot\left(\sum_{n=1}^{19}\cot^{-1}\left(1+\sum_{p=1}^{n}2p\right)\right)$ is

(a) $\frac{23}{22}$ (b) $\frac{22}{23}$ (c) $\frac{19}{21}$ (d) $\frac{21}{19}$

(Main 10th Jan 2nd Shift 2019)

72. All x satisfying the inequality $(\cot^{-1}x)^2 - 7(\cot^{-1}x) + 10 > 0$, lie in the interval

(a) $(-\infty, \cot 5) \cup (\cot 4, \cot 2)$
(b) $(\cot 5, \cot 4)$
(c) $(-\infty, \cot 5) \cup (\cot 2, \infty)$
(d) $(\cot 2, \infty)$ *(Main 11th Jan 2nd Shift 2019)*

73. Considering only the principal values of inverse functions, the set

$$A=\left\{x\geq 0 : \tan^{-1}(2x)+\tan^{-1}(3x)=\frac{\pi}{4}\right\}$$

(a) contains two elements
(b) contains more than two elements
(c) is an empty set
(d) is a singleton *(Main 12th Jan 1st Shift 2019)*

74. The value of $\tan^{-1}\left[\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right]$, $|x|<\frac{1}{2}, x\neq 0,$ is equal to

(a) $\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x^2$ (b) $\frac{\pi}{4}+\frac{1}{2}\cos^{-1}x^2$

(c) $\frac{\pi}{4}-\cos^{-1}x^2$ (d) $\frac{\pi}{4}+\cos^{-1}x^2$

(Main Online 2017)

75. A value of x satisfying the equation $\sin[\cot^{-1}(1 + x)] = \cos[\tan^{-1}x]$, is

(a) 1/2 (b) 0 (c) -1 (d) $-1/2$

(Main Online 2017)

76. Let $\tan^{-1}y=\tan^{-1}x+\tan^{-1}\left(\frac{2x}{1-x^2}\right)$, where $|x|<\frac{1}{\sqrt{3}}$. Then a value of y is

(a) $\frac{3x-x^3}{1+3x^2}$ (b) $\frac{3x+x^3}{1+3x^2}$

(c) $\frac{3x-x^3}{1-3x^2}$ (d) $\frac{3x+x^3}{1-3x^2}$ *(Main 2015)*

77. If $f(x)=2\tan^{-1}x+\sin^{-1}\left(\frac{2x}{1+x^2}\right), x>1$, then $f(5)$ is equal to

(a) $\pi/2$ (b) π

(c) $4\tan^{-1}(5)$ (d) $\tan^{-1}\left(\frac{65}{156}\right)$

(Main Online 2015)

78. If x, y, z are in A.P. and $\tan^{-1}x$, $\tan^{-1}y$ and $\tan^{-1}z$ are also in A.P., then

(a) $2x = 3y = 6z$ (b) $6x = 3y = 2z$
(c) $6x = 4y = 3z$ (d) $x = y = z$ *(Main 2013)*

79. The value of $\cot\left\{\sum_{n=1}^{23}\cot^{-1}\left(1+\sum_{k=1}^{n}2k\right)\right\}$ is

(a) $\frac{23}{25}$ (b) $\frac{25}{23}$ (c) $\frac{23}{24}$ (d) $\frac{24}{23}$

(Advanced 2013)

80. The value of $\cot\left(\operatorname{cosec}^{-1}\frac{5}{3}+\tan^{-1}\frac{2}{3}\right)$ is

(a) $\frac{5}{17}$ (b) $\frac{6}{17}$ (c) $\frac{3}{17}$ (d) $\frac{4}{17}$

(AIEEE 2008)

81. If $0 < x < 1$, then

$$\sqrt{1+x^2}\left[\left\{x\cos\left(\cot^{-1}x\right)+\sin\left(\cot^{-1}x\right)\right\}^2-1\right]^{1/2}=$$

(a) $\frac{x}{\sqrt{1+x^2}}$ (b) x

(c) $x\sqrt{1+x^2}$ (d) $\sqrt{1+x^2}$ *(IIT-JEE 2008)*

82. If $\sin^{-1}\left(\frac{x}{5}\right)+\operatorname{cosec}^{-1}\left(\frac{5}{4}\right)=\frac{\pi}{2}$, then the value of x is

(a) 4 (b) 5 (c) 1 (d) 3

(AIEEE 2007)

83. The largest interval lying in $\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$ for which the function, $f(x)=4^{-x^2}+\cos^{-1}\left(\frac{x}{2}-1\right)+\log(\cos x)$ is defined, is

(a) $\left[-\frac{\pi}{4},\frac{\pi}{2}\right)$ (b) $\left[0,\frac{\pi}{2}\right)$

(c) $[0, \pi]$ (d) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

(AIEEE 2007)

84. If $\cos^{-1}x-\cos^{-1}\frac{y}{2} = \alpha$, then $4x^2 - 4xy\cos\alpha + y^2$ is equal to

(a) 4 (b) $2\sin 2\alpha$
(c) $-4\sin^2\alpha$ (d) $4\sin^2\alpha$ *(AIEEE 2005)*

85. The value of x for which $\sin(\cot^{-1}(1+x)) = \cos(\tan^{-1}x)$ is

(a) $\frac{1}{2}$ (b) 1 (c) 0 (d) $-\frac{1}{2}$

(IIT-JEE 2004)

86. The trigonometric equation $\sin^{-1}x = 2\sin^{-1}a$, has a solution for

(a) all real values (b) $|a| \le \frac{1}{\sqrt{2}}$

(c) $|a| \ge \frac{1}{\sqrt{2}}$ (d) $\frac{1}{2} < |a| < \frac{1}{\sqrt{2}}$

(AIEEE 2003)

87. If $\cot^{-1}\left[(\cos\alpha)^{\frac{1}{2}}\right] + \tan^{-1}\left[(\cos\alpha)^{\frac{1}{2}}\right] = x$, then $\sin x =$

(a) 1 (b) $\cot^2(\alpha/2)$

(c) $\tan\alpha$ (d) $\cot(\alpha/2)$

(AIEEE 2002)

88. If $\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} \ldots\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \ldots\right)$

$= \frac{\pi}{2}$ for $0 < |x| < \sqrt{2}$, then x equals

(a) $\frac{1}{2}$ (b) 1 (c) $-\frac{1}{2}$ (d) -1

(IIT-JEE 2001)

89. The no. of real solutions of

$\tan^{-1}\sqrt{x(x+1)} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$ is

(a) zero (b) one (c) two (d) infinite

(IIT-JEE 1999)

90. If we consider only the principal values of the inverse trigonometric functions then the value of

$\tan\left(\cos^{-1}\frac{1}{5\sqrt{2}} - \sin^{-1}\frac{4}{\sqrt{17}}\right)$ is

(a) $\frac{\sqrt{29}}{3}$ (b) $\frac{29}{3}$ (c) $\frac{\sqrt{3}}{29}$ (d) $\frac{3}{29}$

(IIT-JEE 1994)

91. The principal value of $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$ is

(a) $\frac{-2\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) $\frac{4\pi}{3}$ (d) $\frac{5\pi}{3}$

(e) none of these

(IIT-JEE 1986)

92. The value of $\tan\left(\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{2}{3}\right)$ is

(a) $\frac{6}{17}$ (b) $\frac{17}{6}$

(c) $\frac{16}{7}$ (d) none of these

(IIT-JEE 1983)

One or More Than One Option(s) Correct

93. For any positive integer n, let $S_n : (0, \infty) \to \mathbb{R}$ be defined by

$$S_n(x) = \sum_{k=1}^{n} \cot^{-1}\left(\frac{1 + k(k+1)x^2}{x}\right),$$

where for any $x \in \mathbb{R}$, $\cot^{-1}(x) \in (0, \pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then which of the following statements is (are) TRUE?

(a) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{1+11x^2}{10x}\right)$, for all $x > 0$

(b) $\lim_{n\to\infty} \cot(S_n(x)) = x$, for all $x > 0$

(c) The equation $S_3(x) = \pi/4$ has a root in $(0, \infty)$

(d) $\tan(S_n(x)) \le 1/2$, for all $n \ge 1$ and $x > 0$

(Advanced 2021)

94. For non-negative integer n, let

$$f(n) = \frac{\sum_{k=0}^{n} \sin\left(\frac{k+1}{n+2}\pi\right)\sin\left(\frac{k+2}{n+2}\pi\right)}{\sum_{k=0}^{n} \sin^2\left(\frac{k+1}{n+2}\pi\right)}$$

Assuming $\cos^{-1}x$ takes values in $[0, \pi]$, which of the following options is/are correct?

(a) $f(4) = \sqrt{3}/2$

(b) If $\alpha = \tan(\cos^{-1} f(6))$, then $\alpha^2 + 2\alpha - 1 = 0$

(c) $\sin(7\cos^{-1} f(5)) = 0$

(d) $\lim_{n\to\infty} f(n) = \frac{1}{2}$

(Advanced 2019)

95. If $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$, where the inverse trigonometric functions take only the principal values, then the correct option(s) is(are)

(a) $\cos\beta > 0$ (b) $\sin\beta < 0$

(c) $\cos(\alpha+\beta) > 0$ (d) $\cos\alpha < 0$

(Advanced 2015)

Numerical / Integer Value

96. If $y = \cos\left(\frac{\pi}{3} + \cos^{-1}\frac{x}{2}\right)$, then $(x-y)^2 + 3y^2$ is equal to

(Main 2nd April 2nd Shift 2025)

97. Let the domain of the function $f(x) = \cos^{-1}\left(\frac{4x+5}{3x-7}\right)$ be $[\alpha, \beta]$ and the domain of $g(x) = \log_2(2 - 6\log_{27}(2x+5))$ be (γ, δ). Then $|7(\alpha+\beta) + 4(\gamma+\delta)|$ is equal to______

(Main 8th April 2nd Shift 2025)

98. If for some α, β; $\alpha \le \beta$, $\alpha + \beta = 8$ and $\sec^2(\tan^{-1}\alpha) + \operatorname{cosec}^2(\cot^{-1}\beta) = 36$, then $\alpha^2 + \beta$ is ________.

(Main 24th Jan 1st Shift 2025)

99. Let $S = \{x : \cos^{-1} x = \pi + \sin^{-1}x + \sin^{-1}(2x + 1)\}$. Then $\sum_{x \in S}(2x-1)^2$ is equal to ________.

(Main 29th Jan 1st Shift 2025)

100. For $n \in N$, if $\cot^{-1}3+\cot^{-1}4+\cot^{-1}5+\cot^{-1}n=\frac{\pi}{4}$, then n is equal to________. *(Main 6th April 1st Shift 2024)*

101. Let the inverse trigonometric functions take principal values. The number of real solutions of the equation $2\sin^{-1} x + 3\cos^{-1} x = \frac{2\pi}{5}$, is ________.

(Main 9th April 2nd Shift 2024)

102. If the domain of the function $f(x) = \sec^{-1}\left(\frac{2x}{5x+3}\right)$ is $[\alpha, \beta) \cup (\gamma, \delta]$, then $|3\alpha + 10(\beta + \gamma) + 21\delta|$ is equal to ________.

(Main 10th April 2nd Shift 2023)

103. If $S=\left\{x\in\mathbb{R}:\sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right)-\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)=\frac{\pi}{4}\right\}$, then $\sum_{x\in S}\left(\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right)-\cos\left(\left(x^2+x+5\right)\pi\right)\right)$ is equal to ________. *(Main 13th April 1st Shift 2023)*

104. For $x \in (-1, 1]$, the number of solutions of the equation $\sin^{-1}x = 2\tan^{-1}x$ is equal to ________.

(Main 13th April 2nd Shift 2023)

105. If the sum of all the solutions of $\tan^{-1}\left(\frac{2x}{1-x^2}\right) + \cot^{-1}\left(\frac{1-x^2}{2x}\right) = \frac{\pi}{3}, -1 < x < 1, x \neq 0$, is $\alpha - \frac{4}{\sqrt{3}}$, then α is equal to ________. *(Main 25th Jan 1st Shift 2023)*

106. Let $\tan^{-1}(x)\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, for $x\in\mathbb{R}$.

Then the number of real solutions of the equation $\sqrt{1+\cos(2x)}=\sqrt{2}\tan^{-1}(\tan x)$ in the set $\left(-\frac{3\pi}{2},-\frac{\pi}{2}\right)\cup\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\cup\left(\frac{\pi}{2},\frac{3\pi}{2}\right)$ is equal to

(Advanced 2023)

107. Let $x = \sin(2\tan^{-1}\alpha)$ and $y=\sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$.

If $S = \{\alpha \in R : y^2 = 1 - x\}$, then $\sum_{\alpha\in S}16\alpha^3$ is equal to ________.

(Main 25th July 2nd Shift 2022)

108. For $k \in R$, let the solutions of the equation $\cos(\sin^{-1}(x\cot(\tan^{-1}(\cos(\sin^{-1}x))))) = k,\ 0 < |x| < \frac{1}{\sqrt{2}}$ be α and β, where the inverse trigonometric functions take only principal values. If the solutions of the equation $x^2 - bx - 5 = 0$ are $\frac{1}{\alpha^2}+\frac{1}{\beta^2}$ and $\frac{\alpha}{\beta}$, then $\frac{b}{k^2}$ is equal to ________.

(Main 27th July 1st Shift 2022)

109. $50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}(2\sqrt{2})\right)$ is equal to ________.

(Main 29th June 1st Shift 2022)

110. Considering only the principal values of the inverse trigonometric functions, the value of

$$\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}+\frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2}+\tan^{-1}\frac{\sqrt{2}}{\pi}$$

is ________. *(Advanced 2022)*

111. $\lim_{n\to\infty}\tan\left\{\sum_{r=1}^{n}\tan^{-1}\left(\frac{1}{1+r+r^2}\right)\right\}$ is equal to ________.

(Main 24th Feb 1st Shift 2021)

112. The value of

$$\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)\right)$$

in the interval $\left[-\frac{\pi}{4},\frac{3\pi}{4}\right]$ equals ________.

(Advanced 2019)

113. The number of real solutions of the equation

$$\sin^{-1}\left(\sum_{i=1}^{\infty}x^{i+1}-x\sum_{i=1}^{\infty}\left(\frac{x}{2}\right)^i\right)=\frac{\pi}{2}-\cos^{-1}\left(\sum_{i=1}^{\infty}\left(-\frac{x}{2}\right)^i-\sum_{i=1}^{\infty}(-x)^i\right)$$

lying in the interval $\left(-\frac{1}{2},\frac{1}{2}\right)$ is ________.

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\cos^{-1}x$ assume values in $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$ and $[0, \pi]$, respectively.) *(Advanced 2018)*

114. Let $f: [0, 4\pi] \to [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x)=\frac{10-x}{10}$ is *(Advanced 2014)*

115. Let $f(\theta)=\sin\left(\tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos 2\theta}}\right)\right)$, where $-\frac{\pi}{4}<\theta<\frac{\pi}{4}$.

Then the value of $\frac{d}{d(\tan\theta)}(f(\theta))$ is *(IIT-JEE 2011)*

Matching List

116. Match list-I with list-II and select the correct answer using the code given below the lists.

	List-I		List-II
(P)	$\left(\frac{1}{y^2}\left(\frac{\cos(\tan^{-1}y)+y\sin(\tan^{-1}y)}{\cot(\sin^{-1}y)+\tan(\sin^{-1}y)}\right)^2+y^4\right)^{1/2}$ takes value	(1)	$\frac{1}{2}\sqrt{\frac{5}{3}}$
(Q)	If $\cos x + \cos y + \cos z = 0 = \sin x + \sin y + \sin z$ then possible values of $\cos\left(\frac{x-y}{2}\right)$ is	(2)	$\sqrt{2}$
(R)	If $\cos\left(\frac{\pi}{4}-x\right)\cos 2x+\sin x\sin 2x\sec x = \cos x\sin 2x\sec x+\cos\left(\frac{\pi}{4}+x\right)\cos 2x$ then possible value of $\sec x$ is	(3)	$\frac{1}{2}$
(S)	If $\cot(\sin^{-1}\sqrt{1-x^2})=\sin(\tan^{-1}(x\sqrt{6}))$, $x\neq 0$ then possible value of x is	(4)	1

Codes :

	P	Q	R	S
(a)	4	3	1	2
(b)	4	3	2	1
(c)	3	4	2	1
(d)	3	4	1	2

(Advanced 2013)

117. Let (x, y) be such that

$$\sin^{-1}(a\,x)+\cos^{-1}(y)+\cos^{-1}(bxy) = \frac{\pi}{2}.$$

Column I	Column II
(i) If $a = 1$ and $b = 0$, then (x, y)	(A) lies on the circle $x^2 + y^2 = 1$
(ii) If $a = 1$ and $b = 1$, then (x, y)	(B) lies on $(x^2 - 1)(y^2 - 1) = 0$
(iii) If $a = 1$ and $b = 2$ then (x, y)	(C) lies on $y = x$
(iv) If $a = 2$ and $b = 2$, then (x, y)	(D) lies on $(4x^2 - 1)(y^2 - 1) = 0$

(IIT-JEE 2007)

Subjective Questions

118. Prove that $\cos(\tan^{-1}(\sin(\cot^{-1}x))) = \sqrt{\frac{x^2+1}{x^2+2}}$. *(IIT-JEE 2002)*

119. Find the value of $\cos(2\cos^{-1}x + \sin^{-1}x)$ at $x=\frac{1}{5}$ where $0\leq\cos^{-1}x\leq\pi$ and $-\frac{\pi}{2}\leq\sin^{-1}x\leq\frac{\pi}{2}$ *(IIT-JEE 1981)*

120. Solve the following equation for x.

$$\tan^{-1}2x+\tan^{-1}3x=\frac{\pi}{4}$$ *(IIT-JEE 1978)*

Fill in the Blanks

121. The greater of the two angles $A=2\tan^{-1}(2\sqrt{2}-1)$ and $B=3\sin^{-1}\left(\frac{1}{3}\right)+\sin^{-1}\left(\frac{3}{5}\right)$ is ...... *(IIT-JEE 1989)*

122. The numerical value of $\tan\left(2\tan^{-1}\frac{1}{5}-\frac{\pi}{4}\right)$ is ______. *(IIT-JEE 1984)*

123. Let a, b, c be three positive real numbers

$$\theta=\tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}}+\tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}}+\tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$$

then $\tan\theta$ equals ______. *(IIT-JEE 1981)*